

EXPERIMENT 28:

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier is set to 'SMO' with the following command: `SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"`.

Test options:

- ☐ Use training set
- ☐ Supplied test set (Set...)
- ☒ Cross-validation (Folds: 10)
- ☐ Percentage split (%: 66)

(Nom) class:

- Start
- Stop

Result list (right-click for options):

- 21:06:53 - trees.J48
- 21:42:02 - trees.J48
- 21:42:45 - trees.J48
- 21:46:09 - functions.SMO (selected)

Classifier output:

Linear Kernel: $K(x, y) = \langle x, y \rangle$

Classifier for classes: good, bad

BinarySMO

Machine linear: showing attribute weights, not support vectors.

```
0.6805 * (normalized) checking_status=<0
0.3347 * (normalized) checking_status=0<=X<200
-0.4616 * (normalized) checking_status=>=200
-0.5537 * (normalized) checking_status=no checking
1.6987 * (normalized) duration
0.5398 * (normalized) credit_history=no credits/all paid
0.6015 * (normalized) credit_history=all paid
-0.109 * (normalized) credit_history=existing paid
-0.3182 * (normalized) credit_history=delayed previously
-0.7141 * (normalized) credit_history=critical/other existing credit
0.5673 * (normalized) purpose=new car
-0.5615 * (normalized) purpose=used car
-0.1464 * (normalized) purpose=furniture/equipment
-0.0798 * (normalized) purpose=radio/tv
0.5456 * (normalized) purpose=domestic appliance
-0 * (normalized) purpose=repairs
0.4441 * (normalized) purpose=education
-0.3951 * (normalized) purpose=retraining
-0.0823 * (normalized) purpose=business
-0.2919 * (normalized) purpose=other
1.1473 * (normalized) credit_score
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier is set to 'CostSensitiveClassifier' with the following command: `CostSensitiveClassifier -N "C:\\Program Files\\Weka-3-8-7\\weka.classifiers.meta\\CostSensitiveClassifier"`.

Test options:

- ☐ Use training set
- ☐ Supplied test set (Set...)
- ☒ Cross-validation (Folds: 10)
- ☐ Percentage split (%: 66)

(Nom) class:

- Start
- Stop

Result list (right-click for options):

- 21:06:53 - trees.J48
- 21:42:02 - trees.J48
- 21:42:45 - trees.J48
- 21:46:09 - functions.SMO

Classifier output:

Linear Kernel: $K(x, y) = \langle x, y \rangle$

Classifier for classes: good, bad

BinarySMO

Machine linear: showing attribute weights, not support vectors.

```
0.6805 * (normalized) checking_status=<0
0.3347 * (normalized) checking_status=0<=X<200
-0.4616 * (normalized) checking_status=>=200
-0.5537 * (normalized) checking_status=no checking
1.6987 * (normalized) duration
0.5398 * (normalized) credit_history=no credits/all paid
0.6015 * (normalized) credit_history=all paid
-0.109 * (normalized) credit_history=existing paid
-0.3182 * (normalized) credit_history=delayed previously
-0.7141 * (normalized) credit_history=critical/other existing credit
0.5673 * (normalized) purpose=new car
-0.5615 * (normalized) purpose=used car
-0.1464 * (normalized) purpose=furniture/equipment
-0.0798 * (normalized) purpose=radio/tv
0.5456 * (normalized) purpose=domestic appliance
-0 * (normalized) purpose=repairs
0.4441 * (normalized) purpose=education
-0.3951 * (normalized) purpose=retraining
-0.0823 * (normalized) purpose=business
-0.2919 * (normalized) purpose=other
1.1473 * (normalized) credit_score
```

weka.gui.GenericObjectEditor dialog:

weka.classifiers.meta.CostSensitiveClassifier

About: A metaclassifier that makes its base classifier cost sensitive.

batchSize: 100

classifier: Choose ZeroR

costMatrix: 1 x 1 cost matrix

costMatrixSource: Load cost matrix on demand

debug: False

doNotCheckCapabilities: False

minimizeExpectedCost: False

numDecimalPlaces: 2

onDemandDirectory: Weka-3-8-7

seed: 1

Buttons: Open... Save... OK Cancel

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **CostSensitiveClassifier -N "C:\\Program Files\\Weka-3-8-7" -S 1 -W weka.classifiers.rules.ZeroR**

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) class

Start Stop

Result list (right-click for options)

- 21:06:53 - trees.J48
- 21:42:02 - trees.J48
- 21:42:45 - trees.J48
- 21:46:09 - functions.SMO
- 21:47:19 - meta.CostSensitiveClassifier**

Classifier output

=== Run information ===

Scheme: weka.classifiers.meta.CostSensitiveClassifier -N "C:\\Program Files\\Weka-3-8-7" -S 1 -W weka.classifiers.rules.ZeroR

Relation: german_credit

Instances: 1000

Attributes: 21

- checking_status
- duration
- credit_history
- purpose
- credit_amount
- savings_status
- employment
- installment_commitment
- personal_status
- other_parties
- residence_since
- property_magnitude
- age
- other_payment_plans
- housing
- existing_credits
- job
- num_dependents
- own_telephone
- foreign_worker
- class

Test mode: 10-fold cross-validation

Status

Problem evaluating classifier

Log x 0